

Immunologic Principles in Kidney Transplantation

Orhan Efe, Karl L. Womer, Leonardo V. Riella

s0010 INTRODUCTION

p0010 In 1954, Joseph Murray performed the first successful human kidney transplant between two identical twins. For decades now, kidney transplantation has been the standard of care for kidney failure patients. Maintenance of intact allograft function requires immunosuppression or immune tolerance in genetically nonidentical individuals. Thus, understanding of the immunologic principles remains cardinal in the management of kidney transplant patients (see [Box 105.1](#) for graft terminology).

p0015 Protection against nonself antigens constitutes a paramount function of the immune system. After encountering a foreign antigen, the innate immune system responds by inducing antigen-nonspecific mechanisms such as phagocytosis and cytokine release. Afterwards, antigen-presenting cells (APC) of the innate immune system carry nonself peptides to T lymphocytes in the peripheral lymphoid organs to activate the adaptive immunity. B lymphocytes capture circulating foreign antigens through their B-cell receptors. Consequently, activation of T and B lymphocytes initiate antigen-specific responses that ultimately lead to direct cellular killing and antibody secretion, respectively.

p0020 Similar cascades of events provoke allorecognition and allograft rejection in transplant patients. Recipient's naive T and B cells constitute a diverse repertoire of receptors for an enormous number of potential foreign antigens, including mismatched donor human leukocyte antigens (HLAs). Recognition of mismatched antigens instigates an immune response against the graft parenchyma and the vasculature, leading to allograft rejection ([Fig. 105.1](#)).

p0025 This chapter reviews the immunologic principles in kidney transplantation by dissecting the elements of the innate and the adaptive immune systems, the role of HLA in transplantation, and the pathophysiology of allorecognition and transplant rejection.

s0015 INNATE IMMUNE SYSTEM

p0030 Innate immunity is the frontline defense of the immune system that is activated by the entrance of foreign antigens such as microorganisms and tissue injury. Innate immunity occurs rapidly, with limited specificity and without memory. Innate responses are driven by cellular elements such as neutrophils, macrophages, dendritic cells (DCs), and natural killer (NK) cells, and by molecular components, such as pathogen-associated pattern recognition receptors (PRRs), complement proteins, and cytokines.

The innate response to an insult starts with local chemical signals. Invasion of a microorganism leads to leaking of certain microbial structures called pathogen-associated molecular patterns (PAMP), such as lipoproteins and lipopolysaccharides. Alternatively, sterile tissue damage, such as because of ischemia, leads to release of endogenous molecules termed damage-associated molecular patterns (DAMP), such as adenosine triphosphate (ATP), calcium, uric acid, and DNA. PRRs of the innate system cells then recognize these molecular signals and their activation induces a cascade of events: (1) cytokine release for chemotaxis, (2) labeling of infected or damaged cells (opsonization), and (3) direct cellular killing. Toll-like receptors (TLRs) are the most well-known PRRs, which involve a family of receptors against microbial epitopes including bacteria, virus, and fungi. Similar insults could also activate the complement system through binding of complement factors to microbial proteins (lectin pathway) or antigen-antibody complexes (classical and alternative pathways) that ultimately provoke chemotaxis (by releasing C3a and C5a) and cell death (by terminal complex formation; C5b-9). Overall, these signals alert the immune system to recruit and activate its defense mechanisms.

During donor organ procurement and transplantation, acute ischemia and reperfusion of allograft induce tissue damage and endothelial cell activation by sparking free radicals and inducing apoptosis, which is known as *ischemia/reperfusion injury* (IRI). DAMPs released from damaged graft cells activate PRRs on innate immune cells, resulting in the secretion of cytokines, such as tumor necrosis factor (TNF), type I interferons, chemokines, interleukin (IL)-1, and IL-6.¹ Neutrophils are the prime cells that mediate microvascular plugging and local tissue destruction in IRI. In the next phase, monocyte/macrophage infiltration contributes to the extension of early injury and to repair.² The complement system fixation could occur through activation of classical, alternative, and/or lectin pathway during IRI. Donor-derived APCs migrate to recipient lymphoid tissue to activate lymphocytes. Ultimate parenchymal and vascular injury along with tissue edema contributes to *delayed graft function*. These events may also trigger adaptive, or antigen-specific, immune responses that can induce allograft rejection and negatively affect long-term graft survival.

Adaptive Immune System

B and T lymphocytes and their products, including immunoglobulins and cytokines, make up the adaptive immunity. The adaptive immune response usually follows the innate system activation. After an immune insult, APCs deliver foreign antigens to *secondary lymph organs* (SLOs), such as lymph nodes and spleen, to present them to naive T cells. B cells

BOX 105.1 Graft Terminology

Autograft (autologous graft): A graft from one part of the body to another. Examples include skin and vascular grafts, such as in extensive burn or coronary bypass surgery. No rejection occurs.

Isograft (isogenic or syngeneic graft): A graft from one member of a species to a genetically identical member of the same species. Examples include grafts between identical twins and between members of the same inbred rodent strain. No rejection typically occurs.

Allograft (allogeneic graft): A graft between nonidentical members of the same species. Examples include grafts between unrelated or related non-identical humans and between members of different inbred rodent strains. Rejection occurs primarily by lymphocytes reactive to alloantigens on the graft (i.e., alloresponse).

Xenografts (xenogeneic grafts): A graft between members of different species. Examples include pig or baboon to human and rat to mouse. Rejection occurs by lymphocytes reactive to xenoantigen on the graft (i.e., xenoreponse).

bind circulating soluble antigens through their B-cell receptor (BCR). Each one of the B and T lymphocyte clones is responsive to only a specific foreign antigen because of the selectivity of their receptors. Activation of B and T cells leads to the development of antigen-specific *memory clones* in the SLOs, which enables a more rapid and robust response during repetitive exposure to the same antigen. Activated *effector T cells* emigrate into the inflamed tissue and respond by direct cellular killing ($CD8^+$ cells) and cytokine release ($CD4^+$ cells). B cells respond by producing immunoglobulins against foreign antigens and releasing cytokines in the SLOs and rarely occupy the tissue parenchyma. These antigen-specific responses play a cardinal role during allograft rejection by recognition of donor's mismatched antigens.

Resolution of Inflammation

After the source of antigen is destroyed, both innate and adaptive immune cells help abrogate the inflammation. When activated, certain TLRs concurrently silence certain genes to create a negative feedback loop. Neutrophils and eosinophils catabolize the inflammatory

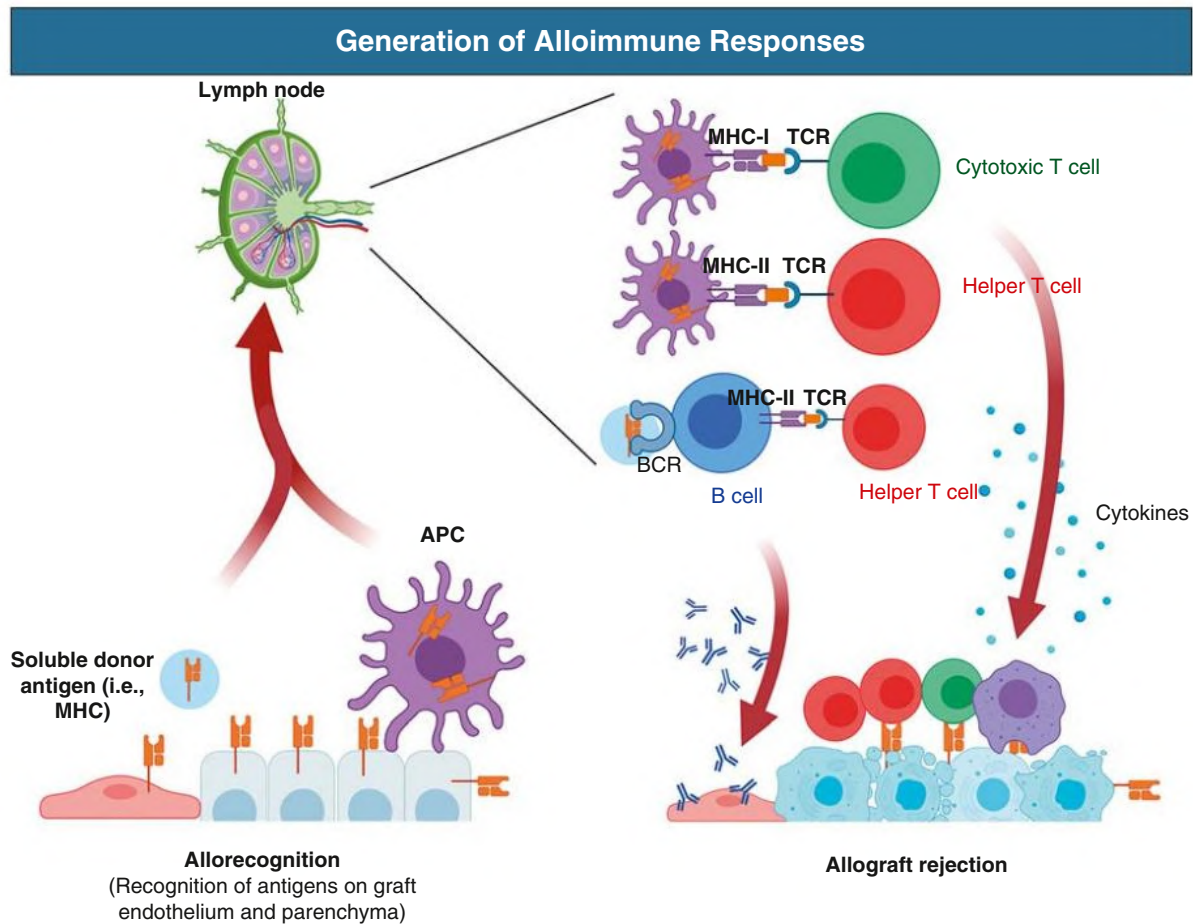


Fig. 105.1 Generation of Alloimmune Responses. Immunologic responses after kidney transplantation represent a series of well-defined stages that result in rejection of the allograft in the absence of exogenous immunosuppression. Injury to allograft tissue such as because of ischemia/reperfusion injury induces innate (antigen-nonspecific) immune responses, which recruit inflammatory cells and initiate adaptive (antigen-specific) immune responses. Donor major histocompatibility complex (MHC) and/or minor proteins that are revealed during tissue injury are taken up by antigen-presenting cells (APCs) of donor (direct pathway) or recipient (indirect pathway) origin. Then, APCs migrate to secondary lymphoid organs (SLO), where they present alloantigen to T cells through MHC structures on their cell surface. After T-cell receptor (TCR) signaling and appropriate costimulation, T cells become activated to produce large amounts of cytokine and undergo clonal expansion. $CD4$ T cells provide help to B cells, $CD8$ T cells, and macrophages for the production of alloantibody, cellular cytotoxicity, and delayed-type hypersensitivity responses, respectively. These effector functions result in destruction of the graft by acute rejection, which may be T-cell and/or antibody mediated. *BCR*, B-cell receptor; *TCR*, T-cell receptor.

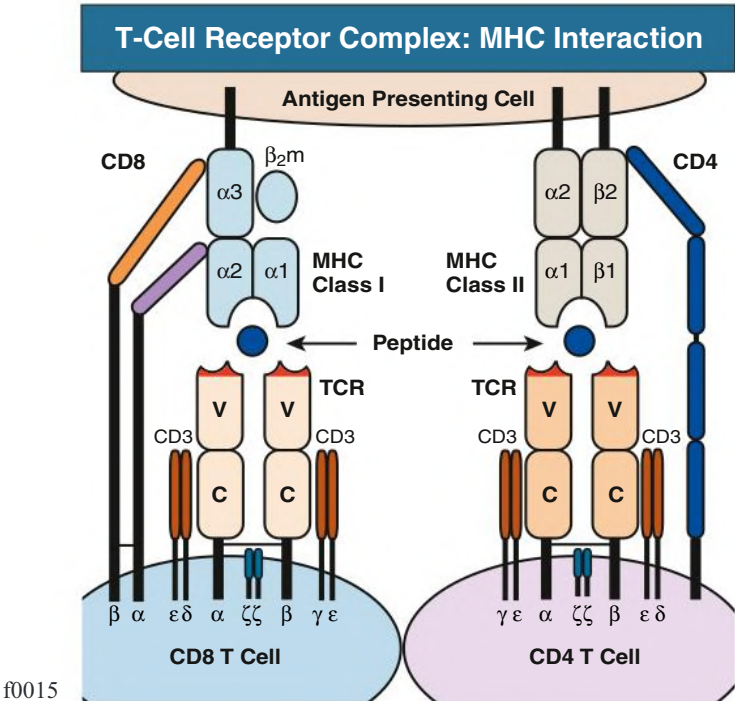


Fig. 105.2 T-Cell Receptor (TCR) Complex: Major Histocompatibility Complex (MHC) Interaction. The formation of a stable immunologic synapse between T cells and antigen-presenting cells is crucial for an effective T-cell activation. Each TCR consists of an α and β chain linked by a disulfide bond. The $\alpha\beta$ heterodimers are similar in structure to the Fab fragment of immunoglobulin molecules (see Fig. 105.4), including variable (V) and constant (C) regions. Diversity in the T-cell repertoire is encoded in the V domains of the α and β chains in three complementarity-determining regions that form the antigen-binding site at the end of the TCR (highlighted in red). CD8 binds to class I MHC molecule on antigen-presenting cells (APCs), whereas CD4 binds to class II MHC molecule on APCs. Triggering of the TCR by antigen initiates a signaling cascade started by the signaling complex made up of CD3 γ , ϵ , and δ chains and the ζ chain homodimer.

mediators such as eicosanoids and cytokines. DCs switch cytokine production from IL-12 to IL-10, favoring the generation of regulatory mechanisms that suppress the function of effector T cells. In the absence of continued cytokine production, T cells lack the necessary growth factors and undergo passive cell death. Activated T cells also undergo activation-induced cell death by the expression of Fas and FasL on their surface, leading to apoptosis.³ Certain regulatory T and B cells also produce antiinflammatory cytokines, such as IL-10 and TGF- β , that suppress immunity.⁴ These mechanisms of immune regulation are crucial for avoiding overactivation and persistence of the immune response once the threat is gone.

selection. *Positive selection* depends on a certain degree of antigen-specific T-cell affinity to self-major histocompatibility complex (MHC) molecules expressed on thymic cortical epithelial cells. This process ensures that mature T cells will interact effectively with self-MHC molecules. *Negative selection* occurs by deletion of T cells with excessively high affinity for self-peptide and MHC, thereby preventing release of high-affinity T cells with autoimmune potential.

Mature T cells expressing their clone-specific TCRs exit the thymus as either CD4 or CD8 T cells. The TCRs of CD4 T cells (also called *T helper cells*) are selected to interact with class II MHC molecules, whereas the TCRs of CD8 T cells interact with class I MHC molecules. Each TCR consists of two different polypeptide chains, termed the TCR α and β chains. In the functional receptor complex, TCR $\alpha\beta$ heterodimers are associated with a complex of four other signaling chains (two ϵ , one δ , one γ) collectively called CD3 (Fig. 105.2). Downstream to TCR, calcineurin acts as an effector through its phosphatase activity by activating NFAT, a transcription factor, leading to IL-2 secretion. This critical molecule for the T-cell activation serves as a target for calcineurin inhibitors in transplantation.

T-Cell Stimulation

Stimulation of T cells are mediated by APCs, which require receptor-ligand interactions and cytokine signaling. In contrast to the BCRs, TCRs do not recognize antigen in its native state but instead recognize a composite ligand of a peptide bound to an MHC molecule. Engagement of the TCR with peptide-bound MHC of an APC is called signal 1. Signal 1, the antigen-specific signal, is effective only in the presence of costimulatory signals, which is termed as signal 2. Costimulation serves as a checkpoint to prevent the activation of self-reactive T cells that escaped negative selection in the thymus. Antigen binding to the TCR in the absence of costimulation fails to activate the T cell and also leads to anergy, in which T cells become refractory to subsequent activation or even undergo apoptosis (programmed cell death). Thus, costimulation removes this inhibition and determines whether a T-cell will proceed with clonal expansion and the development of effector functions. Costimulatory molecules can provide both positive and negative signals to T cells (Fig. 105.3). It is the integration of both positive and negative costimulatory signals during and after initial T-cell activation, dictated by their temporal and spatial expression patterns, that ultimately determines the fate and functional status of the T-cell response.⁶ For an effective and strong T-cell stimulation, cytokine secretion is also required, serving as the signal 3. IL-2 is a cardinal cytokine for T-cell activation and proliferation, by acting in an autocrine or paracrine fashion. IL-2 activation, in turn, triggers another pathway mediated in part through the protein mammalian target of rapamycin (mTOR). New proteins are then translated, allowing the cell to progress from the G1 phase to the S phase of the cell cycle, resulting in proliferation. Table 105.1 lists selected cytokines involved in allograft rejection, their sources, and their effects.

Memory T Cell Formation

Encountering a new foreign antigen transforms part of the naive T cells into memory T cells in SLOs. The principle of immunologic memory is that the immune response to a previously encountered antigen is swifter and more effective than the initial response. Furthermore, there is an increase in antigen-specific T-cell frequency after exposure to a given antigen. *Effector memory T cells* are specialized for quickly entering the inflamed tissues because they can rapidly mature into effector T cells and secrete larger amounts of cytokines independent of costimulatory signals. *Central memory T cells* likely remain in the SLOs and do not produce as much cytokine release.⁷ Memory cells are believed to persist after an initial immune

CELLS OF THE IMMUNE SYSTEM

T Lymphocytes

As a cellular component of adaptive immunity, T cells have a pivotal role in orchestrating the antigen-specific responses by exerting regulatory and effector functions. During T-cell ontogeny in embryogenesis, multilineage bone marrow precursors migrate to the thymus. In the thymus, T-cell receptor (TCR) genes undergo rearrangement to ensure that a diverse repertoire of T cells exists to respond to the enormous number of potential foreign antigens, including foreign HLAs, which play a crucial role in alloresponses.⁵ The mature T-cell repertoire is determined in the thymus by two processes, positive and negative

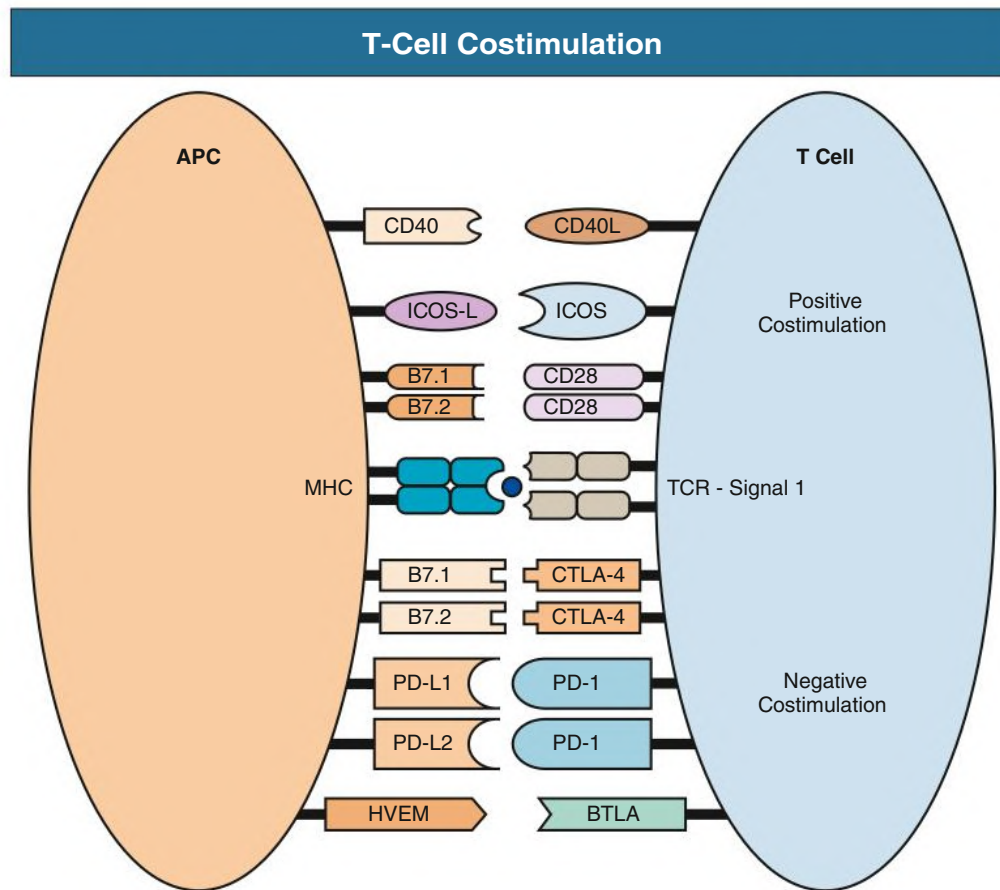


Fig. 105.3 T-Cell Costimulation. Binding of the TCR:CD3 complex to peptide:MHC on antigen-presenting cells (APCs) delivers a signal that can induce the clonal expansion of naive T cells only when the appropriate costimulatory signal is delivered (Signal 2). CD28 and its ligands, B7.1 (CD80) and B7.2 (CD86), are the best-characterized costimulatory molecules that induce clonal expansion of naive CD4 T helper cells. Once activated, T cells express increased levels of CTLA-4 (CD152). CTLA-4 has an affinity 10 to 20 times greater than that of CD28 for B7 molecules and thus binds most or all of the B7 molecules, effectively shutting down the proliferative phase of the response. B- and T-lymphocyte attenuator (BTLA) and programmed death-1 (PD-1) are two other coinhibitory molecules on T cells. Activated T cells express proteins that contribute to sustaining or modifying the costimulatory signal to drive clonal expansion and differentiation. The inducible costimulatory molecule (ICOS) is a CD28 homologue, but unlike CD28 is not constitutively expressed on naive T cells. Rather, ICOS is induced only after T-cell activation. Engagement of ICOS by B7H enhances T-cell proliferation, cytokine production, and survival. Binding of CD40 by CD40L, which is upregulated by CD28 signaling, transmits activating signals to the T cell but also activates the APCs to secrete proinflammatory molecules and to express B7 molecules, thus stimulating further T-cell proliferation. *HVEM*, Member of the tumor necrosis factor receptor superfamily; *MHC*, Major histocompatibility complex; *TCR*, T-cell receptor.

response through expression of the antiapoptotic genes *Bcl-2* and *Bcl-xL*, which are induced primarily by IL-2 and CD28 stimulation. Long-term memory cell survival is likely a function of periodic interactions with self-MHC:peptide complexes on APCs. Memory T cells constitute an important risk factor for allograft rejection in patients with history of HLA sensitization through pregnancy, blood transfusion, or prior transplant.

s0050 CD4 T Cells

p0075 CD4 T cells have both effector and regulatory functions, mostly through secreting specific cytokines. After prolonged stimulation, CD4 T cells express their signature cytokines, probably depending on the local environment, nature of the antigen, and type and activation status of the APC. These different T helper (Th) subsets, each with unique transcription factor and cytokine signatures, are referred to as Th1, Th2, Th17, and Tfh (follicular help) populations.

The Th1 clones produce IL-2, IFN- γ , and lymphotoxin, which are growth and maturation factors for cytotoxic T lymphocytes (especially IL-2) and macrophages (particularly IFN- γ). T helper 17 (Th17) cells produce proinflammatory cytokines, including IL-17, IL-21, and IL-22.⁸ Th1, in particular, plays a dominant role in acute cellular rejection. Th2 clones secrete TGF- β , IL-4, IL-5, and IL-10, which facilitate B cell activation and class switching. Th cells are found in lymph nodes and are also important for B cell maturation. Thus, these two cell types contribute to antibody-mediated rejection by facilitating antibody production.

In addition to T cells that promote immune responses, there are CD4 T-cell subpopulations that control immune responses termed *regulatory T cells* (Tregs). Tregs are responsible for peripheral tolerance by suppressing autoimmunity and contribute to the resolution of inflammation by secreting inhibitory cytokines such as TGF- β , IL-10, and IL-35. The most well-characterized Treg is known for its

t0010 **TABLE 105.1 Cytokines Involved in Allograft Rejection**

Cytokine	Source	Biologic Activity
IL-1	Macrophages, DCs, ECs, NK cells	Proinflammatory, adhesion molecule expression on ECs, NK cell function
IL-2	Activated T cells	T-cell proliferation, CTL and NK cell function, Treg maintenance, immunoglobulin production by B cells, AICD of activated T cells
IL-4	Activated T cells	Activated T and B cell proliferation, Th2 differentiation, allergic responses, MHC II upregulation on B cells
IL-6	T cells, macrophages, ECs	Proinflammatory and anti-inflammatory, acute-phase responses
IL-10	T cells, Tregs, Bregs, macrophages, DCs	Anti-inflammatory, suppression of APC function, NK cell inhibition
IL-12	Macrophages, DCs	Proinflammatory, Th1 differentiation, NK cell and CTL activity, IFN- γ and TNF- α production by NK and T cells
IL-15	Epithelial cells, stromal cells, macrophages	NK cell proliferation, T-cell proliferation, memory T-cell survival
IL-17	T cells	Proinflammatory and allergic responses, Th17 function, cytokine production from many cell types
IFN- γ	Activated Th1 cells, CTLs, DCs, NK cells	MHC expression by EC, macrophage function, Th1 differentiation, Th2 suppression, adhesion and binding of T cells to ECs, NK cell activity
TGF- β	T and B cells, Tregs, Bregs, macrophages, platelets	Anti-inflammatory, wound healing, fibrosis
TNF- α	Macrophages, T and B cells, ECs, NK cells	Proinflammatory, acute phase responses, cytotoxicity

AICD, Activation-induced cell death; Breg, regulatory B cells; CTL, cytotoxic T lymphocytes; DC, dendritic cells; EC, endothelial cells; IFN- γ , interferon- γ ; IL, interleukin; NK, natural killer; TGF- β , transforming growth factor- β ; Th, T helper (cell); TNF- α , tumor necrosis factor- α ; Treg, regulatory T cells.

high expression of Foxp3 and IL-2 receptor alpha chain, CD25. Treg-expanding strategies, such as low-dose IL-2 or infusion of ex vivo expanded Tregs, have shown promising results in patients with autoimmune disorders and in transplantation.^{9–11}

s0055 **CD8 T Cells**

p0090 CD8 T cells are effector cells with functions of direct cellular killing and they are called *cytotoxic T lymphocytes* (CTLs) in their activated form. CTLs possess two mechanisms to kill target cells that require cell-to-cell contact.¹² First, they release perforin and granzyme B from specialized lytic granules. The granzyme B–perforin complex enters the cell through the mannose 6-phosphate receptor, and, after internalization, perforin allows granzyme B to enter the cell by pore formation on the vesicle surface to induce programmed cell death through apoptosis.¹³ The second mechanism is mediated by Fas/Fas ligand (FasL) interactions. Fas (CD95) on the target cell surface is activated by FasL on CTLs to induce apoptosis. Both pathways induce apoptosis through activation of the caspase cascade in target cells.¹⁴ CD8 CTLs also release several cytokines that exert direct cytotoxic effects, including IFN- γ , TNF- α , and TNF- β .¹⁵ Current studies delineated the presence of CD4 CTLs that own similar cytolytic features to CD8 CTLs.¹⁶

s0060 **B Lymphocytes**

p0095 B cells protect extracellular spaces of the body by producing antibodies and contribute to antibody-mediated allograft rejection. B-cell development starts in the primary lymphoid organs including fetal liver (until birth) and bone marrow from common lymphoid progenitor cells. From the very early stages of development, B cells constitute a BCR, which contains an antigen specific immunoglobulin (Ig)M and/or IgD component and associated antigen nonspecific Ig- α and - β proteins (Fig. 105.4). In primary lymphoid organs, autoreactive B cells undergo either *receptor editing* to eliminate self-reactivity or *negative selection* by apoptosis.

p0100 Survivor naive B cells emigrate into SLOs where the majority die and the rest will switch into mature B cells to become either memory cells or plasma cells. In the SLOs, B cells continue to mature by *class switching* to produce IgA, IgE, or IgG apart from IgM and IgD and *somatic hypermutation* of variable domains of BCR to further increase their affinity to a given antigen. The first could occur either in a T-cell-dependent or independent manner that leads to differentiation into the

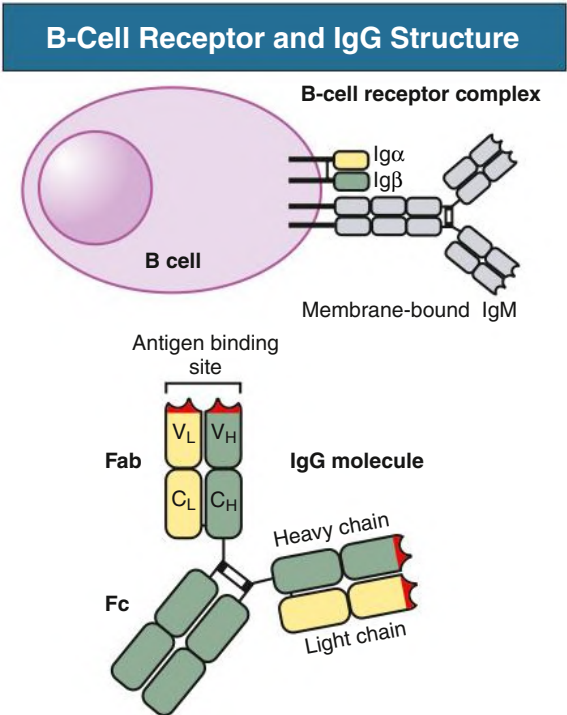


Fig. 105.4 B-Cell Receptor and Immunoglobulin G (IgG) Structure. B-cell surface membrane-bound immunoglobulin M (IgM) associated with antigen-nonspecific signaling molecules, Ig α and Ig β , forms the B-cell receptor complex. The IgG molecule is made up of four polypeptide chains, involving two identical light (L) chains (yellow) and two identical heavy (H) chains (green). Each of the four chains has a variable (V) region at its amino terminus (Fab portion), which contributes to the antigen-binding site, and a constant (C) region (Fc portion), which determines the isotype. The V domains contain hypervariable regions that determine antigen specificity called complementarity-determining regions (highlighted in red).

short-lived plasma cells that remain in the SLOs. The latter only ensue in the presence of T helper cells in the germinal centers, which evolve higher affinity B cells into memory cells and plasma cells generating a long-lived humoral immunity. BAFF and APRIL, which are survival

f0025

T-Cell– and Antibody-Mediated Rejection

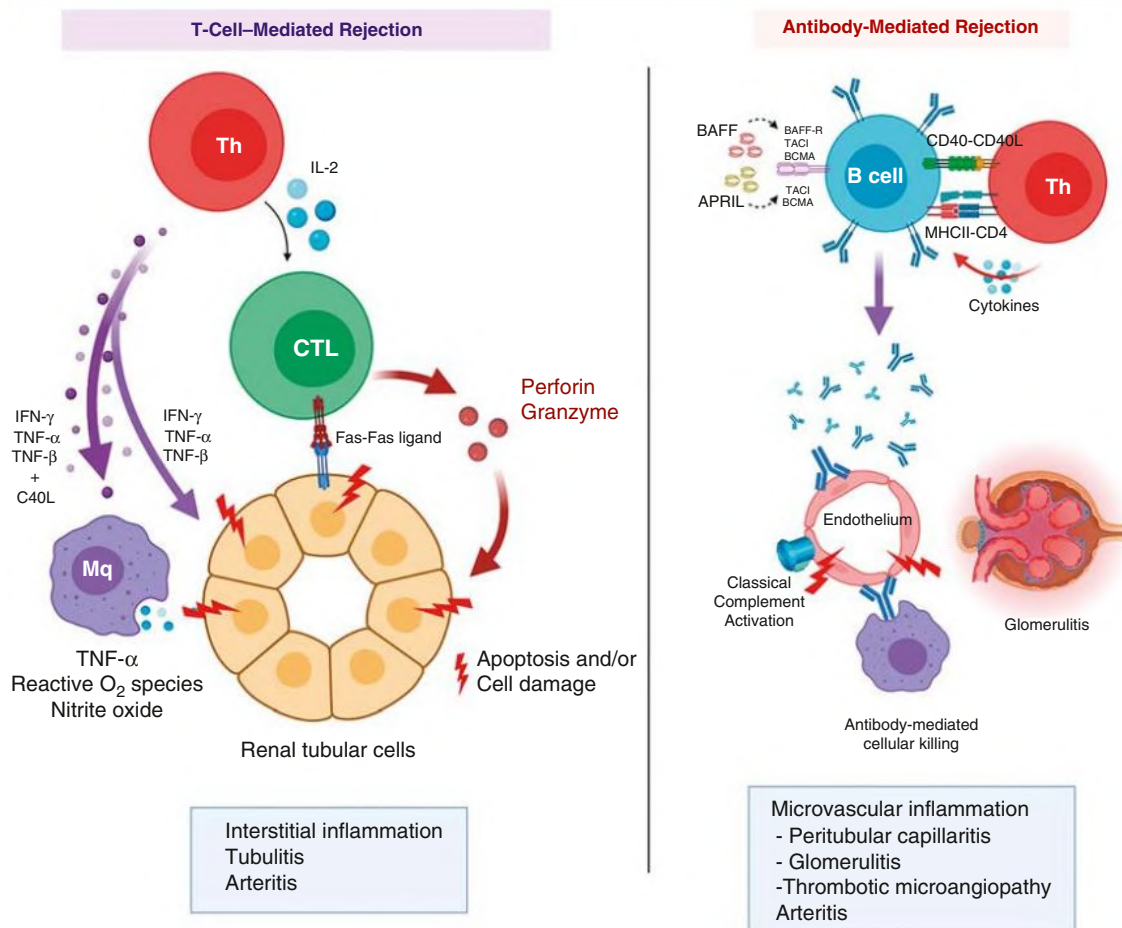


Fig. 105.5 T-Cell-Mediated and Antibody-Mediated Rejection. CD4 T cells induce epithelial and endothelial cell damage directly by secretion of cytokines but also indirectly by activation of cytotoxic T lymphocytes (CTLs) and macrophages. CTLs may cause apoptosis by releasing cytolytic granules containing granzymes and perforin or by exposure of Fas ligand (FasL) on the T-cell surface. Macrophages induce local tissue damage through secretion of cytokines, oxygen species, and nitric oxide (delayed-type hypersensitivity response). CD4 T cells secrete cytokines, which induce upregulation of major histocompatibility complex (MHC) molecules on epithelial and endothelial cells. CD4 T cells also provide help to B cells for production of alloantibody by engagement of CD40L and production of cytokines. BAFF and APRIL induce B-cell development and survival. Antigraft antibody is usually directed at MHC molecules, and it leads to injury either through the activation of the complement pathway or by recruiting innate cytotoxic cells through the Fc receptor binding to the antibody. Damaged endothelial cells secrete factors that activate coagulation systems and result in formation of microthrombi. Exposure of tubular cells and glomerular capillary loops to alloimmunity over time results in interstitial fibrosis, tubular atrophy, and glomerulosclerosis. BAFF-R, BAFF receptor; BCMA, B-cell maturing antigen; CTL, cytotoxic T lymphocytes; IFN, interferon; Mq, macrophages; TACI, T cell activator and calcium modulating ligand interactor; TNF, tumor necrosis factor; Th, T helper cell.

stimulating proteins for B cells, contribute to optimal development of B cells (Fig. 105.5). After their generation, plasma cells locate into bone marrow where a niche for longevity exists.

B cells express CD-19 and CD-20 starting from pro-B cell and pre-B cell phase, respectively, up to memory B cells. These markers are lost during differentiation into plasma cells and replaced by high expression of CD-38. Given differential expression of these surface markers, rituximab (a monoclonal antibody against CD-20 used to treat antibody-mediated rejection) targets a broad range of B cell stages but excludes primitive B cells and plasma cells.

There are three effector functions of B-lymphocytes: (1) antibody production, (2) antigen presentation to T cells, and (3) cytokine

production. Immunoglobulins function in several different ways once bound to their target antigen: fixation of complement, opsonization for phagocytosis by Fc receptor (FcR)-positive cells (including B cells, NK cells, macrophages, and neutrophils), opsonization for cell lysis by the cells capable of antibody-dependent cellular cytotoxicity (NK cells, macrophages, neutrophils, and eosinophils), and induction of eosinophil degranulation. The five major classes of antibodies are IgM, IgD, IgG, IgA, and IgE. IgG is by far the most abundant immunoglobulin. The IgG molecule is made up of two identical light (L) and two identical heavy (H) chains that form a flexible Y-shaped structure (see Fig. 105.4). Variable (V) regions of each of the four chains contain the antigen specific part (Fab portion), and constant (C) regions determine

HLA Inheritance

f0035

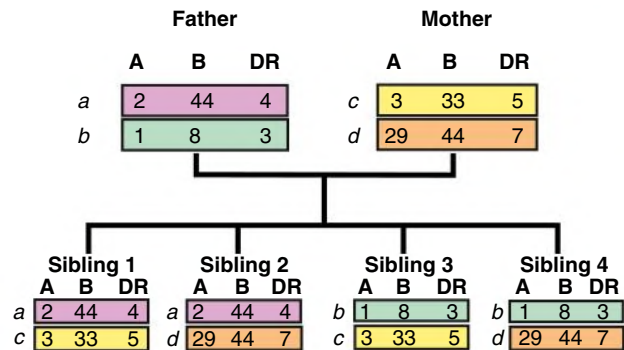


Fig. 105.6 Example of Human Leukocyte Antigen (HLA) Inheritance. HLA antigens are inherited and expressed in a mendelian codominant fashion, whereby one copy of each HLA gene called a haplotype (e.g., a, b, c, d) is inherited from each parent. Efforts are made to match both class I (HLA-A and -B) and class II (HLA-DR) antigens.

the isotype (Fc portion). The V regions of a given antibody contain hypervariable segments that determine antigen specificity by forming a surface complementary to the antigen referred to as *complementarity-determining regions* (CDRs).

s0065 **ANTIGEN-PRESENTING CELLS**

p0115 APCs are specialized cells with capability of activating T cells. DCs, macrophages, and B lymphocytes are considered “professional APCs,” although DCs are the most potent. Professional APCs engulf target antigens by phagocytosis, migrate into T cell areas of lymphoid organs, and present processed antigen on MHC I molecule to CD8 T cells or on MHC II molecule to CD4 T cells (signal 1). As previously explained, costimulation and cytokine signals (signal 2 and 3) are also essential for a successful T-cell stimulation. In the allograft after transplantation, APCs (mostly DCs) of either donor or host origin take up the donor MHC proteins and present to recipient T cells.

p0120 Two most important APCs during transplant rejection are macrophages and DCs. *Macrophages* are recruited to the graft in response to proinflammatory cytokines, such as IL-1 and IL-6. Th1 and CD8 T cells activate macrophages by producing IFN-γ or binding of CD40L-CD40. As an effector function, macrophages mediate cytotoxicity by producing TNF-α, oxygen radicals, and nitric oxide. Activated macrophages also can produce IL-12, which directs the differentiation of activated naive CD4 T cells into Th1 effector cells. On the other hand, macrophage activation is inhibited by cytokines such as transforming growth factor-β (TGF-β) and IL-10, many of which are produced by Th2 and regulatory T and B cells. DCs are also part of the mononuclear phagocyte system. As an effector function, DCs release IL-12 and type I and III interferons to induce Th1 differentiation and CD8 T-cell activation.^{17,18} Lastly, *local endothelium* and *parenchyma* could also serve as in situ APCs when they are activated.¹⁹

s0070 **Natural Killer Cells**

p0125 NK cells are a subset of peripheral lymphocytes that share developmental and functional features with CD8 T lymphocytes.²⁰ Unlike T or B cells, NK cells do not possess an antigen-specific cell surface receptor that is generated by gene recombination. Instead, NK cells use receptors that recognize the loss of HLA class I molecules on susceptible targets. Thus, NK cells can recognize when self-MHC class I is absent, the so-called “missing self,” which triggers their activation.

Peripheral NK cells are mature, do not require costimulation or differentiation as with T cells, and immediately release cytotoxic granules and inflammatory cytokines such as TNF-α and IFN-γ on detection of relevant targets. Given the strong cytolytic function and potential for autoreactivity, NK cell activity is tightly regulated. Mechanisms of activation include cytokines, binding of antibody to FcRs, and binding of ligands to activating receptors. In kidney transplantation, NK cells contribute to allograft rejection through antibody-mediated cytotoxicity in the presence of donor-specific antibodies (DSAs) and missing self-mechanism on the allograft cells with HLA class I mismatch.²¹

MAJOR HISTOCOMPATIBILITY COMPLEX AND ALLORECOGNITION

The MHC, which encodes HLAs, are the most polymorphic molecules in humans and are made up of closely linked series of genes at the short arm of chromosome 6. MHC genes are inherited in a mendelian codominant fashion (Fig. 105.6) and divided into three regions: class I, II, and III. Class I and II HLAs play a key role in antigen presentation and self-recognition. The key antigens encoded by class I MHC genes include HLA-A, -B, and -C, and the class II gene products include HLA-DP, -DQ, and -DR.

The class I and II proteins share structural homology but are functionally different²² (Fig. 105.7). MHC I is designed to sample intracellular proteins to detect tumors or intracellular pathogens, such as virus and intracellular bacteria. It is expressed on all nucleated cells. MHC I-peptide complex is recognized by CD8 T cells, which leads to destruction of target cells. The class II MHC system is designed to sample extracellular proteins that have been taken up by APCs to present to CD4 T helper cells. Class II proteins are expressed by only limited cell types including “professional” APCs (DCs, B cells, macrophages, and Langerhans cells) and by activated parenchymal and endothelial cells. Importantly, both class I and II MHC expression are prompted on allograft cells by IFN-γ in synergy with other cytokines during transplant rejection, which leads to a sustained inflammation. Lastly, certain APCs employ *cross-presentation* by taking up, processing, and presenting extracellular antigen on their class I MHC molecules to CD8 T cells.²³ This mechanism is used for generation of immunity against tumors and viruses that do not infect APCs.

In transplantation, recipient T cells encounter alloantigen by either direct, semidirect, or indirect pathways of *allorecognition* depending on

s0075

p0130

p0135

p0140

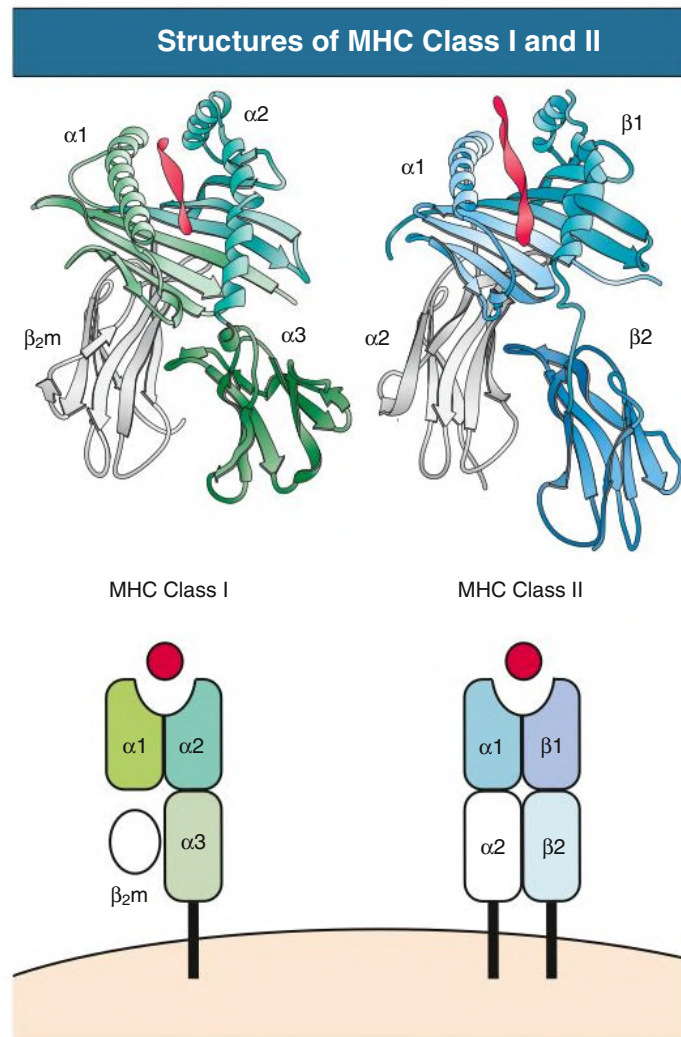


Fig. 105.7 Structures of Major Histocompatibility Complex (MHC) Classes I and II. MHC class I is composed of a heavy chain divided into $\alpha 1$, $\alpha 2$, and $\alpha 3$ domains noncovalently associated with β_2 -microglobulin (β_2m). The $\alpha 1$ and $\alpha 2$ domains each form a long α helix and β sheet to make up the floor and walls of the peptide-binding groove (peptide indicated by red ribbon). MHC class II is a dimer composed of α and β chains. Each chain is divided into two domains, with $\alpha 1$ and $\beta 1$ domains forming the two α helices and β -pleated sheet that surround the peptide-binding groove.

recipient's or donor's APC involvement (Fig. 105.8). In the *direct pathway*, donor APCs present their own peptides, most importantly MHC proteins, on their intact MHCs to recipient T cells. The direct pathway may be more active early after transplantation, when a large number of donor APCs are present in the allograft. The direct pathway also occurs later when recipient T cells directly interact with intact donor MHC molecules on the surface of endothelium or parenchymal cells. The *indirect pathway* is the physiologic mechanism of foreign antigen recognition in which recipient APCs present donor antigens to recipient T cells.²⁴ Lastly, the *semidirect pathway* involves the transfer of intact donor MHC:peptide complexes from donor APCs or from the graft cells to the surface of recipient APCs to be delivered to recipient T cells, a process called *cross-dressing* or *cell nibbling*.²⁵

which predisposes to allograft rejection. Originally, HLA typing was performed by mixing donor or recipient's lymphocytes with sera from multiparous women or persons who had received blood transfusions. Currently, HLA typing is performed by molecular biology techniques including polymerase chain reaction (PCR) sequencing or next-generation sequencing, which allows more precise analysis. Although complete allele matching between donor and recipient would minimize the risk of sensitization against donor kidney, this also reduces likelihood of receiving a transplant offer. Hence, studies compared impact of matching different HLA types on graft survival. Currently, HLA-A, -B, and -DR alleles are used in HLA-matching algorithms, but this could show variation between centers.^{27,28}

In addition to HLA matching, the patient should be screened for preexisting anti-HLA antibodies (DSA), which are associated with antibody-mediated rejection and poor graft survival. Currently, single antigen bead assay is the preferred method for DSA testing. In this method, patient serum is added to multiple different HLA-coated beads. Binding of antibodies from patient's serum is detected with a secondary antibody. Potential recipient's antibody profile is used to

s0080 Human Leukocyte Antigen Typing and Transplantation

p0145 MHC class I and II genes are highly polymorphic in the regions that encode the peptide-binding groove, leading to thousands of different HLA types (see www.ebi.ac.uk/imgt/hla).²⁶ Mismatched donor HLAs are regarded as foreign antigens by the recipient's immune system,

Direct, Semidirect, and Indirect Antigen Presentation

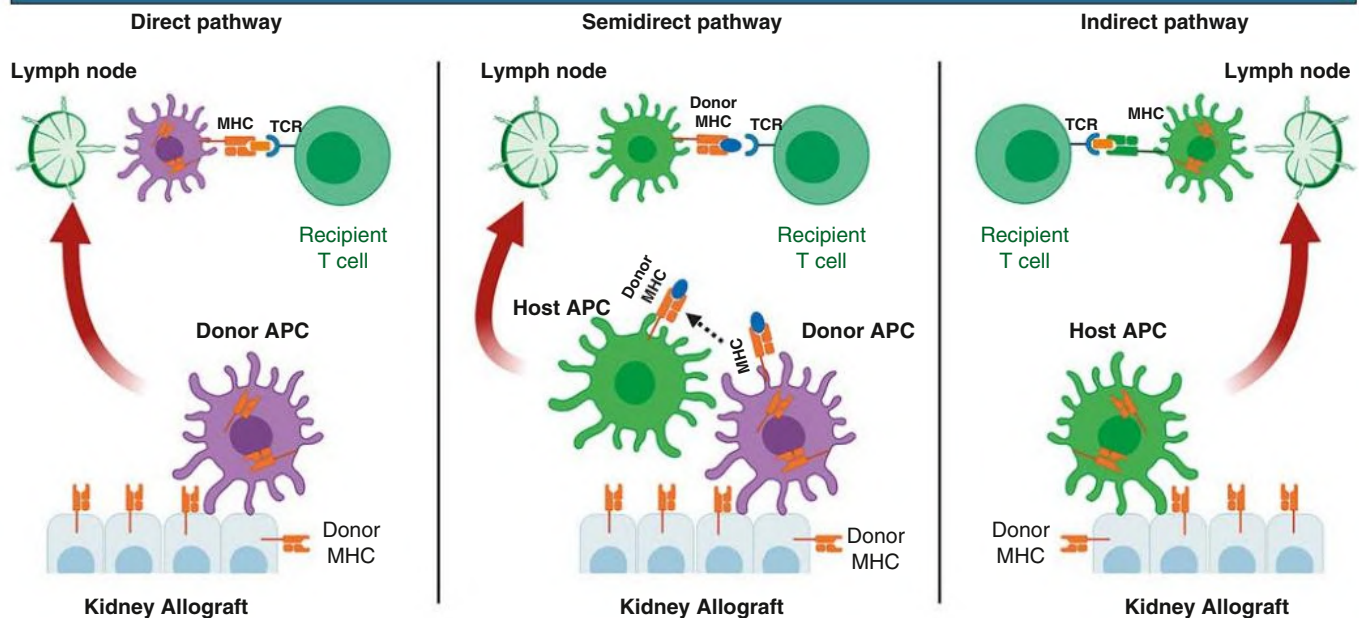


Fig. 105.8 Direct, Semidirect, and Indirect Antigen Presentation. In direct allorecognition, donor antigen (shown in red) is presented to recipient T cells as a peptide in the context of intact donor major histocompatibility complex (MHC) molecules on the surface of donor antigen-presenting cells (APCs). In semidirect allorecognition, donor APCs transfer their own peptide-MHC complexes to the surface of recipient APCs (without intracellular processing) that deliver them to recipient T cells. In indirect allorecognition, the donor antigen is processed by recipient APCs and presented as a peptide in the context of recipient MHC molecules.

determine *calculated panel-reactive antibody*, which gives the percentage of potentially positive-cross matches among the HLA types in a donor pool.²⁹ Defining the presence of specific anti-HLA antibodies may also allow us to perform a virtual crossmatch, which entails verifying the presence of any anti-HLA antibodies that may react against a specific donor HLA antigen. This approach could help exclude potential recipients who may have a positive cross-match test against a specific donor without actually performing the cross-match assay.

Non-Major Histocompatibility Antigens

Minor histocompatibility antigens are normal proteins that are polymorphic within a given species. Even when a transplant donor and recipient are identical with regard to MHC genes, amino acid differences in these minor proteins can lead to rejection. Minor antigens are encoded by a large number of chromosomes and are presented only as peptides in the context of recipient MHC (indirect allorecognition). Minor antigens are responsible for the need for immunosuppression after donation between HLA matched but nonidentical twin siblings. The prototypic minor histocompatibility antigen, the male or H-Y antigen, is derived from a group of proteins encoded on the Y chromosome. Alloresponses to this antigen are responsible for the rejection of male mouse skin grafts by otherwise identical female recipients and may explain observations of reduced long-term graft survival in human male-to-female donations. MHC I-related chain A (MICA) antigens are endothelial surface glycoproteins with functions related to innate immunity. Exposure to allogeneic MICA during transplantation was shown to elicit antibody formation.³⁰ ABO blood group glycolipids expressed on endothelial and red blood cells and angiotension-1 receptors are other notable non-MHC antigens.³¹

Allograft Rejection

Allograft rejection is defined as tissue injury produced by the effector mechanisms of the alloimmune response, leading to deterioration of graft function.³² There are two main types of rejection: T-cell-mediated or cellular rejection (TCMR) and antibody-mediated rejection (AMR). Both types of rejection can be early or late, fulminant or indolent, and isolated or concomitant and can share pathologic features on biopsy.¹ Kidney allograft biopsies have been evaluated according to the Banff classification, which guides clinicians in treating the different allograft pathologies.³³

Allograft rejection is caused by several cellular elements of the immune system, including T cells, B cells, plasma cells, macrophages, NK cells, and neutrophils. Although there are a variety of target cells in the graft, endothelial and tubular cells are particularly affected by these mediators. T cells serve as the main effectors and regulators of the alloimmune response, and macrophages serve as effectors but also aid in the removal of apoptotic cells. B cells and plasma cells serve in the production of alloantibodies, and neutrophils and NK cells may cause significant damage, particularly during AMR through antibody-mediated cell toxicity.^{21,34}

Acute T-Cell-Mediated Rejection

Acute TCMR starts with the phagocytosis of donor antigens by either the donor or recipient's APCs, such as macrophages and DCs (see Fig. 105.8). Afterwards, APCs migrate into SLOs and activate naive or memory T cell clones, which then migrate into the allograft. Inflammation in the allograft microenvironment grows in time by chemotaxis of further macrophages and T cells. Although a small number of B cells may also be present in the TCMR infiltrate, these are likely recruited nonspecifically.³⁵ The first pathologic event in TCMR

is *interstitial inflammation*, and this is followed by T cell infiltration of the tubules, causing *tubulitis*. Thus, interstitial nephritis and tubulitis constitute two core features of TCMR (see Fig. 105.5). Similar inflammation can take place in the arteries causing *endarteritis* in a minority of cases.

p0175 Deterioration of kidney function during TCMR correlates with tubulitis and arterial inflammation (endothelialitis), which is less common. CTLs can kill target cells by release of cytotoxic molecules (perforin, granzyme B, and granulysin) or engagement of Fas on target cells by FasL. CD4 T-helper cells, especially Th1, release numerous cytokines that affect the alloimmune response. They stimulate macrophages to produce nitric oxide, reactive oxygen species, and TNF- α .³⁶ T-cell cytokines also act directly on parenchymal cells or endothelium. TNF- α and TNF- β exert local cytotoxic effects on graft endothelial cells and tubular cells through TNF receptors. IFN- γ , the prototypical Th1 cytokine, induces MHC class II expression on endothelium and MHC class I expression on vascular endothelial cells, epithelial cells, and parenchymal cells in the graft promoting the allorecognition. Production of cytokines such as IL-2 by CD4 T cells provides help for the generation of CTLs from their CD8 T-cell precursors. CD4 T cells may themselves become cytolytic T cells, by mechanisms similar to those used by CD8 CTLs. CD4 T cells also provide help to B cells to enhance their production of alloantibodies. Finally, Foxp3 + CD4 Tregs that are responsible for peripheral immune regulation are concentrated in the tubules during TCMR as a counterregulatory mechanism.

s0100 Acute Antibody-Mediated Rejection

p0180 Acute AMR may develop because of either preexisting antibodies in presensitized patients or de novo antibody formation against donor HLA antigens, which together are called DSAs. In de novo antibody formation, donor antigens are shed into the bloodstream and interact with B cells in SLOs, which induce differentiation of naive B cells into memory B cells and plasma cells (see Fig. 105.5). Although AMR typically occurs with antibody response against the donor HLA antigens, it can also occur to non-HLA antigens (minor antigens). *Hyperacute rejection*, which is most commonly because of the presence of preformed antidonor HLA antibody in sensitized recipients, is a form of AMR that can result in immediate graft failure. This type of AMR is rare now given current anti-HLA testing and crossmatching practices employed by transplant centers.

p0185 Alloantibody-antigen binding mediates graft damage through different mechanisms including opsonization for phagocytosis, opsonization for antibody-dependent cellular cytotoxicity, and fixation of complement.³⁷ The kidney typically shows an accumulation of NK cells, neutrophils, and monocytes in peritubular capillaries (PTCs) and glomeruli (*microvascular inflammation*), although the infiltrate can be quite sparse.³⁸ As a result of antibody-mediated damage, endothelium release von Willebrand factor and proinflammatory cytokines, leading to platelet aggregation and microthrombi and the aggravation of inflammation.^{39,40} *Endarteritis* is another pathologic finding observed in AMR, which could also be seen in TCMR in a lesser degree.⁴¹ Tubulitis is generally minimal, unless a component of TCMR is present.

p0190 Only the *classical pathway* has been shown to participate in acute or chronic AMR. Binding of C1 to alloantibody-antigen complex initiates classical complement cascade. C4b and its inactive fragment C4d are products of the classical pathway. Both contain an occult sulfhydryl group that forms a covalent thioester bond with nearby proteins in the tissue. No functional role for C4d is known, but it remains in the tissue for several days after immunoglobulin and C1 have been released. C4d deposition is strongly associated with circulating antibody to donor HLA class I or II antigens and is currently the best single marker of

complement-fixing circulating antibodies to the endothelium. C3a and C5a drive chemoattraction of neutrophils and macrophages.⁴² The membrane attack complex, C5b-9, causes lysis of endothelial cells.

The current diagnostic criteria for acute AMR include the histologic evidence of acute tissue injury (in the absence of other possible causes), immunopathologic evidence of current/recent antibody interaction with vascular endothelium (e.g., C4d staining or microvascular inflammation), and the evidence of circulating antibody reactive to the donor HLA or minor antigens.

Chronic Rejection

Late allograft dysfunction is because of both alloimmune mechanisms (i.e., chronic rejection) and alloimmune-independent mechanisms, including hypertension, calcineurin inhibitor toxicity, and recurrent disease.⁴³ Chronic rejection may occur by either cellular or humoral mechanisms, or both.

Chronic active AMR is diagnosed in the presence of histologic evidence for chronic injury including transplant glomerulopathy, peritubular capillaropathy, or arteriopathy; immunopathologic evidence of antibody action (e.g., C4d staining), and evidence of circulating antibody reactive to the donor (HLA or minor antigens).³⁸ *Transplant glomerulopathy* is defined by the widespread duplication or multilamination of the glomerular basement membrane (GBM), sometimes accompanied by mesangial expansion and accumulation of mononuclear cells in glomerular capillaries. Majority of cases of transplant glomerulopathy are associated with circulating DSAs, and approximately 30% of these cases have C4d deposition in the PTCs.⁴⁴ Absence of C4d deposition in the remainder of cases may represent intermittent antibody involvement or complement-independent antibody-mediated injury. In *peritubular capillaropathy*, PTCs display multilamination under electron microscopy. Repeated episodes of antibody-mediated injury to the endothelium may result in repair mechanisms characterized by duplication of the basement membrane. What leads to episodic antibody injury is unknown, but fluctuating DSA levels are observed in some patients. Lastly, *transplant arteriopathy* is characterized by arterial intimal fibrosis without cellular elements, which ultimately narrows vascular lumen, reducing blood flow to the glomeruli and tubules.

Chronic active TCMR is diagnosed when the histology shows interstitial fibrosis along with at least moderate interstitial inflammation, concurrent tubulitis, and the presence of inflammation in at least 25% of the cortex. An alternative diagnostic criterion is the presence of allograft arteriopathy with arterial intimal fibrosis, mononuclear cell infiltration in the fibrosis, and formation of neointima.⁴⁵

TRANSPLANTATION TOLERANCE

Transplantation tolerance is a state characterized by the absence of a destructive immune response in the recipient toward a well-functioning donor allograft, with a fully intact immune system and no exogenous immunosuppression.^{46,47} As with self-tolerance, transplantation tolerance is achieved through control of T-cell reactivity by both central and peripheral mechanisms. *Central tolerance* involves thymic deletional mechanisms that eliminate T cells with reactivity against self-antigens (or donor antigens in the case of transplantation tolerance). In experimental transplant models, central tolerance is achieved by elimination of the preexisting mature T-cell population by irradiation and/or cytotoxic agents, followed by infusion of donor hematopoietic progenitor cells. Reconstituted donor antigen “reeducates” the thymus to delete developing T cells with antidonor reactivity, leading to a state of *chimerism*, in which both donor and recipient cells coexist. Although adult humans lack a thymus, this was translated into the clinical setting with concurrent bone marrow and kidney transplantation

along with transient immunosuppression, which led to promising outcomes in case series.⁴⁸

p0220 *Peripheral tolerance* mechanisms include deletion, anergy, and regulation. With self-tolerance, these mechanisms prevent deleterious autoimmune responses from T cells that escape central deletion. Tregs play a pivotal role in peripheral tolerance. Various approaches to induce peripheral transplant tolerance with the goal of inhibiting deleterious alloimmune responses are currently under investigation, including costimulatory blockade, pharmacologic manipulation of DCs, and the induction of Tregs. Clinical studies indicated that the

selective stimulation of Tregs through IL-2 receptors could induce peripheral tolerance and allow the reduction of the immunosuppression in autoimmunity and potentially transplant recipients.^{9,11} Most recently, the infusion of ex vivo expanded Tregs in the kidney transplant patients enabled reduction of immunosuppression safely and effectively.¹⁰ A hurdle to the development of clinical tolerance strategies is the lack of reproducible immune monitoring assays for tolerance, which would determine when immunosuppressive medications could be safely withdrawn. Although transplantation tolerance is not yet a clinical reality, progress is being made.^{49,50}

SELF-ASSESSMENT QUESTIONS

- o0010 1. Which one of the following costimulation molecules on APCs has an inhibitory function on T cells?
- o0015 A. B7.1
- o0020 B. B7.2
- o0025 C. CD40
- o0030 D. ICOS-L
- o0035 E. PD-L1
- o0040 2. Which B-cell development stages are not targeted by rituximab (anti-CD20 antibody)?
- o0045 A. Pro-B cell
- o0050 B. Pre-B cell
- o0055 C. Memory B cell
- o0060 D. Plasma cells
- o0065 E. A and D
- o0070 3. Which of the below options includes the pathologic findings of acute T-cell-mediated rejection?
- o0075 A. Peritubular capillaritis, tubulitis, interstitial inflammation
- o0080 B. Glomerulitis, peritubular capillaritis, arteritis
- o0085 C. Tubulitis, interstitial inflammation, arteritis
- o0090 D. Glomerulitis, tubulitis, interstitial inflammation

- E. Interstitial inflammation, tubulitis, thrombotic microangiopathy o0095
4. Which one of the following statements is *not* true regarding HLA genes? o0100
- A. HLA genes are inherited in a mendelian codominant fashion. o0105
- B. Siblings from the same parents are predicted to have a 50% chance of having zero HLA mismatches. o0110
- C. In kidney transplantation, efforts are made to avoid the presence of preformed antibodies against donor HLA antigens. o0115
- D. One haplotype of HLA genes is inherited from each parent. o0120
5. Which one of the following statements is *not* true regarding alloantigen presentation to T cells? o0125
- A. T cells recognize alloantigen via direct, indirect, and semidirect pathways. o0130
- B. Class II MHC molecules are recognized by CD4 T cells. o0135
- C. The MHC class I system is designed to present extracellular proteins taken up and processed by antigen-presenting cells. o0140
- D. T cells recognize alloantigen as peptides presented by MHC molecules. o0145

1. Wood KJ, Goto R. Mechanisms of rejection: current perspectives. *Transplantation*. 2012;93:1–10.
2. Swaminathan S, Griffin MD. First responders: understanding monocyte-lineage traffic in the acutely injured kidney. *Kidney Int*. 2008;74:1509–1511.
3. Wood KJ, Bushell A, Hester J. Regulatory immune cells in transplantation. *Nat Rev Immunol*. 2012;12:417–430.
4. Serhan CN, Brain SD, Buckley CD, et al. Resolution of inflammation: state of the art, definitions and terms. *FASEB J*. 2007;21(2):325–332.
5. Schwartz RS. Shattuck lecture: diversity of the immune repertoire and immunoregulation. *N Engl J Med*. 2003;348:1017–1026.
6. Li XC, Rothstein DM, Sayegh MH. Costimulatory pathways in transplantation: challenges and new developments. *Immunol Rev*. 2009;229:271–293.
7. Willinger T, Freeman T, Hasegawa H, et al. Molecular signatures distinguish human central memory from effector memory CD8 T cell subsets. *J Immunol*. 2005;175:5895–5903.
8. Ouyang W, Kolls JK, Zheng Y. The biological functions of T helper 17 cell effector cytokines in inflammation. *Immunity*. 2008;28:454–467.
9. Saadoun D, et al. Regulatory T-cell responses to low-dose interleukin-2 in HCV-induced vasculitis. *N Engl J Med*. 2011;365(22):2067–2077.
10. Sawitzki B, et al. Regulatory cell therapy in kidney transplantation (The ONE Study): a harmonised design and analysis of seven non-randomised, single-arm, phase 1/2A trials. *Lancet*. 2020;395(10237):1627–1639.
11. Koreth J, et al. Interleukin-2 and regulatory T cells in graft-versus-host disease. *N Engl J Med*. 2011;365(22):2055–2066.
12. Barry M, Bleackley RC. Cytotoxic T lymphocytes: all roads lead to death. *Nat Rev Immunol*. 2002;2:401–409.
13. Buzza MS, Bird PI. Extracellular granzymes: current perspectives. *Biol Chem*. 2006;387:827–837.
14. Barry M, Chris Bleackley R. Cytotoxic T lymphocytes: all roads lead to death. *Nat Rev Immunol*. 2002;2(6):401–409.
15. Al-Lamki RS, Wang J, Skepper JN, et al. Expression of tumor necrosis factor receptors in normal kidney and rejecting renal transplants. *Lab Invest*. 2001;81:1503–1515.
16. Takeuchi A, Saito T. CD4 CTL, a cytotoxic subset of CD4+ T cells, their differentiation and function. *Front Immunol*. 2017;8:194.
17. Collin M, McGovern N, Haniffa M. Human dendritic cell subsets. *Immunology*. 2013;140(1):22–30.
18. Macatonia SE, et al. Dendritic cells produce IL-12 and direct the development of Th1 cells from naive CD4+ T cells. *J Immunol*. 1995;154(10):5071–5079.
19. Valujskikh A, Lakkis FG. Remembrance of things past: memory T cells and transplant rejection. *Immunol Rev*. 2003;196:65–74.
20. Young NT. Immunobiology of natural killer lymphocytes in transplantation. *Transplantation*. 2004;78:1–6.
21. Koenig A, et al. Missing self-induced activation of NK cells combines with non-complement-fixing donor-specific antibodies to accelerate kidney transplant loss in chronic antibody-mediated rejection. *J Am Soc Nephrol*. 2020.
22. Jones EY. MHC class I and class II structures. *Curr Opin Immunol*. 1997;9:75–79.
23. Bevan MJ. Cross-priming. *Nat Immunol*. 2006;7:363–365.
24. Gokmen MR, Lombardi G, Lechler RI. The importance of the indirect pathway of allorecognition in clinical transplantation. *Curr Opin Immunol*. 2008;20:568–574.
25. Alegre ML, Lakkis FG, Morelli AE. Antigen presentation in transplantation. *Trends Immunol*. 2016;37:831–843.
26. Gao X, Nelson GW, Karacki P, et al. Effect of a single amino acid change in MHC class I molecules on the rate of progression to AIDS. *N Engl J Med*. 2001;344:1668–1675.
27. Opelz G. Correlation of HLA matching with kidney graft survival in patients with or without cyclosporine treatment. *Transplantation*. 1985;40:240–243.
28. Doxiadis II, et al. Simpler and equitable allocation of kidneys from postmortem donors primarily based on full HLA-DR compatibility. *Transplantation*. 2007;83:1207–1213.
29. Sypek M, et al. HLA epitope matching in kidney transplantation: an overview for the general nephrologist. *Am J Kidney Dis*. 2018;71(5):720–731.
30. Zou Y, Stastny P, Susal C, et al. Antibodies against MICA antigens and kidney-transplant rejection. *N Engl J Med*. 2007;357:1293–1300.
31. Farouk S, Zhang Z, Madhav C, Menon. Non-HLA donor–recipient mismatches in kidney transplantation—a stone left unturned. *Am J Transplant*. 2020;20(1):19–24.
32. Ponticelli C. The mechanisms of acute transplant rejection revisited. *J Nephrol*. 2012;25:150–158.
33. Roufosse C, Simmonds N, Clahsen-van Groningen M, et al. 2018 reference guide to the Banff classification of renal allograft Pathology. *Transplantation*. 2018;102(11):1795–1814.
34. Scozzi D, et al. The role of neutrophils in transplanted organs. *Am J Transplant*. 2017;17(2):328–335.
35. Halloran PF. T cell–mediated rejection of kidney transplants: a personal viewpoint. *Am J Transplant*. 2010;10(5):1126–1134.
36. Halloran PF. T cell–mediated rejection of kidney transplants: a personal viewpoint. *Am J Transplant*. 2010;10:1126–1134.
37. Rahimi S, Qian Z, Layton J, et al. Non-complement- and complement-activating antibodies synergize to cause rejection of cardiac allografts. *Am J Transplant*. 2004;4:326–334.
38. Colvin RB. Antibody-mediated renal allograft rejection: diagnosis and pathogenesis. *J Am Soc Nephrol*. 2007;18:1046–1056.
39. Ota H, Fox-Talbot K, Hu W, et al. Terminal complement components mediate release of von Willebrand factor and adhesion of platelets in arteries of allografts. *Transplantation*. 2005;79:276–281.
40. Camerer E, Huang W, Coughlin SR. Tissue factor- and factor X-dependent activation of protease-activated receptor 2 by factor VIIa. *Proc Natl Acad Sci USA*. 2000;97:5255–5260.
41. Lefaucheur C, et al. Antibody-mediated vascular rejection of kidney allografts: a population-based study. *Lancet*. 2013;381(9863):313–319.
42. Colvin RB, Smith RN. Antibody-mediated organ-allograft rejection. *Nat Rev Immunol*. 2005;5:807–817.
43. Womer KL, Vella JP, Sayegh MH. Chronic allograft dysfunction: mechanisms and new approaches to therapy. *Semin Nephrol*. 2000;20:126–147.
44. Sis B, Campbell PM, Mueller T, et al. Transplant glomerulopathy, late antibody-mediated rejection and the ABCD tetrad in kidney allograft biopsies for cause. *Am J Transplant*. 2007;7:1743–1752.
45. Haas M, et al. The Banff 2017 Kidney Meeting Report: revised diagnostic criteria for chronic active T cell–mediated rejection, antibody–mediated rejection, and prospects for integrative endpoints for next–generation clinical trials. *Am J Transplant*. 2018;18(2):293–307.
46. Salama AD, Womer KL, Sayegh MH. Clinical transplantation tolerance: many rivers to cross. *J Immunol*. 2007;178:5419–5423.
47. Newell KA. Clinical transplantation tolerance. *Semin Immunopathol*. 2011;33:91–104.
48. Kawai T, et al. HLA-mismatched renal transplantation without maintenance immunosuppression. *N Engl J Med*. 2008;358(4):353–361.
49. Fudaba Y, Spitzer TR, Shaffer J, et al. Myeloma responses and tolerance following combined kidney and nonmyeloablative marrow transplantation: in vivo and in vitro analyses. *Am J Transplant*. 2006;6:2121–2133.
50. Kawai T, Cosimi AB, Spitzer TR, et al. HLA-mismatched renal transplantation without maintenance immunosuppression. *N Engl J Med*. 2008;358:353–361.